1 Question

Given an array of integers that are greater than zero as a parameter, write a function called **splitIndex** that checks if it is possible to split this array in two (without changing the order of the elements), such that the sum of the two resulting arrays is the same. The method returns the splitting index (starting index of the second array) if it is possible to split the array; -1 otherwise.

```
public static int splitIndex(int[] A)
```

```
A =  {5, 2, 3}
Returns 1.
A = {2, 3, 2, 1, 1, 1, 2, 1, 1}
Returns 3.
A = {1, 1, 3}
Returns -1.
A = {1, 1, 3, 1}
Returns -1.
```

2 Question

Write a method named **repeatedSequence** that accepts two arrays of integers $a1$ and $a2$ as parameters and returns $true$ if $a2$ is composed entirely of repetitions of $a1$ and $false$ otherwise. For example, if $a1$ stores the elements $\{2, 1, 3\}$ and $a2$ stores the elements $\{2, 1, 3, 2, 1, 3, 2, 1, 3\}$, the method would return $true$. If the length of $a2$ is not a multiple of the length of $a1$, your method should return $false$. You may assume that both arrays passed to your method will have a length of at least 1. The following examples show some calls to your method and their expected results:

```
int[] a1 = {23};
int[] a2 = {23, 23, 23, 23};
repeatedSequence(a1, a2) returns true

int[] a3 = {5, 6, 7, 8};
int[] a4 = {5, 6, 7, 8};
repeatedSequence(a3, a4) returns true

int[] a5 = {5, 6};
int[] a6 = {5, 6, 7, 5, 6, 5};
repeatedSequence(a5, a6) returns false
```

3 Question

Write a method named **outliers** that reads N integers from a file and prints the integers that are two standard deviation above or below the mean of those N numbers into another file. Standard deviation of N numbers is calculated as:

$$s = \sqrt{\frac{\sum_{i=1}^{N}(x_i - m)^2}{N-1}}$$

. Your method should use the following methods

- a method named **readfile** to read N integers from an input file and returns an array
- a method named **average** to find the average of N numbers stored in an array
- a method named **stdev** to find the standard deviation of N numbers stored in an array

which must also be implemented.

4 Question

Write a non-recursive method **aBeforeB**

```
int aBeforeB(String s)
```

which returns the number of 'a' characters preceding a 'b' character in the string *s*.

5 Question

Write a recursive method **sumOfTriples**

```
int sumOfTriples(int N)
```

which calculates

$$1 \times 2 \times 3 + 2 \times 3 \times 4 + 3 \times 4 \times 5 + \ldots + (N-1) \times N \times (N+1)$$

```
sumOfTriples(2) -> 6
sumOfTriples(4) -> 90
```

6 Question

Write a recursive method **noX**, which takes a string as a parameter and returns recursively a new string where all the 'x' chars have been removed.

```
String noX(String str)

noX("xaxb") -> "ab"
noX("abc") -> "abc"
noX("xx") -> ""
```